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An exploratory study of university librarians' perceptions on causes and deterrents of plagiarism: a Pakistani perspective

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Abstract

Purpose – The role of academic libraries in dissemination of information and facilitating research of academic community is well-established world over. However, a significant role of libraries in ensuring fair use of information to promote an ethical research culture at the higher education institutes is also flourishing. This study aims to explore the perceived causes of unethical use of literature by university scholars and deterrents to plagiarism in universities of Pakistan.

Design/methodology/approach – The design of the study was qualitative, and interviews were conducted on nine university library heads in Pakistan who were officially administrating anti-plagiarism (Turnitin) software service at their universities and were providing "similarity index certification" of PhD and MPhil research work through the software. They were purposively selected for the interviews because of their direct interaction with the researchers and faculty on matters of information searching as well as its ethical use. The study is based on PhD thesis work of the first author.

Findings – The findings of the study depicted that conceptual unawareness of plagiarism, lack of information skills, socio-cultural factors and weak policies were the prime reasons for unethical research practices among graduate scholars. Participants were positive that a role of university library in combating plagiarism through provision of anti-plagiarism software and guidance could prove more beneficial for both the researchers and libraries.

Originality/value – Multiple studies have been conducted, in and out of south Asian region, to gather the views of the students and faculty on the matters of academic misconduct. However, no study was found presenting the views of the other stakeholders in promotion of academic integrity. To the best of the authors' knowledge, this study is first of its kind in South Asian context and provided observation-based views of a group of university library heads who are also administrating anti-plagiarism service in their universities on the causes and deterrents to plagiarism.

Keywords Pakistan, Libraries, Universities, Plagiarism, Academic integrity, Research culture, Library heads, Prevention

Paper type Research paper

1. Background of the study

The word "plagiarism" has its origin in the Latin word "plagiarius" meaning "kidnapper" and was first used by a Latin poet Martial to refer to literary theft in his poems (Seo, 2009). In simple words, plagiarism is essentially the act of committing a fraud. It is the process of presenting the intellectual work or idea of someone else as one's own. There are traces of unethical research practices in academic institutions from various parts of the world with varying frequencies from country to country and institution to institution and the region of

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South Asia is not an exception. The Web surfing of the south Asian literature regarding academic integrity issues indicates that scholars from all countries of this region not only highlighted the problems and incidents of plagiarism (Miah *et al.*, 2011; Jayaraman, 2012) in their countries but also stressed measures to control the problems (Adhikari, 2010; Khanzai, 2015).

Plagiarism has become a sensitive issue in the growing academic research culture of Pakistan and literature has frequently reported instances of both intentional and unintentional plagiarism at various universities. Regardless of the type of plagiarism, it is always a source of embarrassment for the institution involved and results in loss of reputation and goodwill in the local and international forums. Furthermore, the production of authentic knowledge through research is a major objective of any university. Therefore, in Pakistan, Higher Education Commission (HEC) established a Quality Assurance Division in 2005, to provide assistance to universities in maintaining ethical standards and quality in research among other things.

The Higher Education Commission Pakistan is the first higher education governing body in South Asia to implement a zero-tolerance plagiarism policy and allocate funds for providing access to anti-plagiarism software to HEC recognized public sector universities in the country. From 2007, it was also mandated that all research work of PhD, MPhil and international publications must be submitted for the anti-plagiarism software check prior to their final submission and focal persons were nominated to administrate the software service in each university. Piracha (2011) further informed that for University of the Punjab, Pakistan the Chief Librarian of its central library (Punjab University Library) was nominated as focal person and designated for the responsibility. Therefore, Punjab University Library was the first academic library of Pakistan which started officially anti-plagiarism services for its university in South Asian region. The role of academic libraries in facilitating research through organization and dissemination of information is well-established world over. However, a significant role of libraries in ensuring fair use of information to promote an ethical research culture at the higher education institutes is flourishing. Therefore, the role of University Libraries in Pakistan in combating plagiarism is also not much described in the available literature so far. Therefore, it appears necessary to investigate the current practices of University Libraries to prevent plagiarism, perceived reasons and deterrents of plagiarism in academic research. Higher Education Governing bodies in South Asian countries, like University Grant Commission of India recommended in 2012 in its five year plan to initiate a plagiarism detection policy for its universities (Bailey, 2013; University Grant Commission India, 2012), followed the footsteps of HEC Pakistan and formulated their anti- plagiarism policies and implemented the use of anti-plagiarism software in their universities. Therefore, the findings of this study may act as a baseline for future research on the issues of plagiarism prevention in South Asia and in revising policies. The study may prove beneficial in creating awareness and motivation in the librarians' professional community regarding their role in promotion of academic integrity and ethical use of information. Furthermore, the results may be useful for any future research conducted on designing of information literacy instruction programs offered by libraries. Multiple studies have been conducted, in and out of south Asian region, to gather the views of the students and faculty on the matters of academic misconduct. However, no study was found presenting the views of the other stakeholders in promotion of academic integrity. This study is first of its kind in South Asian context and provided observation-based views of a group of anti-plagiarism service administrators in Pakistani universities on the causes and deterrents to plagiarism.

2. Literature review

Researchers reported various factors that cause academic misconduct by students including ignorance, peer pressure, the role of technology and cultural differences.

2.1 Ignorance

Lack of awareness regarding research ethical practices among students has been reported by many scholars as a major contributory factor in accelerating plagiarism incidents. Harris (2001) stated that a significant ratio of plagiarism incidents occurred due to ignorance with the ethical standards of information usage and report writing. Pecorari (2003) confirms that "Most students will use sources inappropriately before they learn how to use them appropriately" (p. 342). Ramzan *et al.* (2012) reported that students in Pakistani universities have a poor level of conceptual understanding of plagiarism. Williams (2005) states that there are three types of plagiarists among academics:

(1) the lazy plagiarist who starts to work on the last dates and falls in the ditch due to time pressure and stress;

(2) the cunning plagiarist who is clear about what plagiarism is, but works hard to avoid detection; and

(3) the accidental plagiarist who emerges as "a consequence of inexperience, poor study skills, local academic norms or some combination thereof. Such students typically insert slabs of unattributed text in their essays and, when challenged, will be either embarrassed by their sloppy referencing or genuinely surprised that they have been challenged at all, claiming ignorance of the system" (pp. 4–5).

2.2 Peer pressure and time management

On one side, poor time management or laziness and on the other side, competitive academic pressure plays an important role in generating stress among the scholars that leads towards intentional plagiarism. Batane (2010) reported that in Botswana University the reason for plagiarism occurrence was simple laziness as only 7% of the students choose skill deficiency as a reason for academic dishonesty and stated, "It is tempting to plagiarize because lecturers give the same assignments year after year" (p. 8). McCabe and Trevino (1993) found the attitude of peers an influential cause of plagiarism among students due to a competitive environment.

Harris (2001) argued that impact of peers behaviour becomes more powerful when there is no penalty for academic misconducts and "sometimes students give excuses that people at role models positions – from parents to politicians do all these unethical deeds and for students, it becomes as learned behaviour" (p. 6). Adhikari (2010) describes that peers behaviour is one among other reasons for a high level of self-plagiarism in medical research in Nepal. She states that mostly research in academics is conducted to publish in peer-reviewed journals as a requirement for an academic appointment or promotion and to gain social and financial benefits.

The finding of the study of Curtis and Popal (2011) negated the perception that all students did cheating to get higher grades. They established that brilliant and hardworking students were found less involved in academic dishonesty practices than the students who were less motivated towards studies. The authors suggested that creating awareness and serious attitude towards academic values would be helpful to reduce unethical academic practices.

2.3 Educational culture

Maxwell *et al.* (2008) explored the influences of educational culture on understanding and perceived seriousness towards plagiarism between Australian and Asian students. They reported that rote learning was still heavily used in Asian academic institutions. They believed that this technique causes "learned plagiarism" among Asian scholars and creates a clash between the Western ideas of "textual ownership" and Eastern practice for learning. The difference in writing skills of the Asian scholars as compared to the Australian scholars might be another reason for a higher ratio of plagiarism incidents among former. Khan (2012) mentioned that in most of the developing countries like Pakistan English language is used as primary academic language. He believed that it is difficult for most of the scholars to write fluently in a foreign language, thus making copy and pasting is an easy option to finish the research task.

The studies indicate that the elements of ignorance, peer pressures, technology and cultural differences play an important role in plagiarism. A few studies conducted to gather cultural perspectives on plagiarism mentioned differences in plagiarism perception, research standards and practices along with ethical values as the prime elements of high ratio of plagiarism incidents among Asian scholars (Hayes and Introna, 2005; Maxwell *et al.*, 2008; Yusof, 2009). However, these studies present the perceptions of students and faculty on the causes of plagiarism. The views of other stakeholders involved in plagiarism prevention practices were not mentioned anywhere.

3. Role of technology and anti-plagiarism software

Role of technology in making plagiarism easy was considered vital by many researchers (Auer and Krupar, 2001; Lampert, 2008; Bartlett and Casselden, 2011). The scholars believed that due to free and quick information access through the web along with cut and paste technique, students were more likely to copy from the Internet than printed resources. In Pakistan, Ramzan *et al.* (2012) believed that internet provided extensive opportunities for unethical use of information due to this free access. However, others believe that students as likely commit plagiarism from printed resources as they do from electronic resources. Ison (2015) establishes empirically that the mean of similarity index (measured through Turnitin software) for doctoral dissertations written before 1994 (mean 14.5%) is higher than in the dissertation written after 2010 (mean 12.3%). Hence, he believes that it is unjustified to declare the use of internet as a major cause of plagiarism. Although he accepted that ratio of extensive copying of unattributed material was more common in the research written after 2010.

Scholars have reported the use of anti-plagiarism software not only as a plagiarism-detecting tool but also as a learning tool for the prevention of plagiarism. They have also discussed the limitations of the method at length. Kloda and Nicholson (2007) indicated that Turnitin has increased awareness against plagiarism on campuses and is used as an alternative method of deterring plagiarism. Similarly, Davis and Carroll (2009) stated that the use of Turnitin in academic institutions had brought two positive effects. Firstly, it increased the awareness level among students about the existence of plagiarism and put a check on intentional plagiarism. Secondly, it has initiated a discussion about the use of alternative methods of combating plagiarism due to its limitations in dealing with both intentional and unintentional plagiarism.

Barrett and Malcolm (2006) stated that anti-plagiarism softwares could be used to educate students in learning the correct style and methods of citations and paraphrasing. Davis and Carroll (2009) conducted a case study on 66 international students of Oxford 26 Brookes University, UK over three years, to investigate the potential use of text matching

software in anti-plagiarism training initiatives targeting students at the formative stages of their academic work. The results of the study showed that students found Turnitin to be helpful in avoiding plagiarism and for – finding bad paraphrasing|| in their assignments. However, the writers also indicated that Turnitin is not able to teach the required standard of paraphrasing to its users. It is something that can only be learned through – repeated practice and extensive reading (p. 67). They also proposed that the approach towards plagiarism should be changed from catch-and-punish to formative learning. Bretag and Mahmud (2009) have reported that despite the various shortcomings of Turnitin – electronic detection in combination with manual analysis, nuanced academic judgement and clear processes provide the means to determine if plagiarism has occurred (p. 1). They concluded as a result their study, that if Turnitin was used within a learning framework it could enhance the students' writing abilities and improve the overall situation of academic integrity on a campus. Therefore, they suggested that anti-plagiarism software should be approached as an aid to a coherent and positive educational approach to academic honesty, rather than as a quick shortcut to stop plagiarism and cheating.

Batane (2010) conducted a pilot study on the students of University of Botswana to determine the impact of Turnitin on the level of plagiarism offences. He discovered that while the fear of Turnitin had decreased the level of plagiarism among students, it had not completely eliminated it. He found that students "still plagiarized even when they knew their papers will be checked through the software" as they believed that by reducing amount of material plagiarized they "could still get away with that kind of cheating since it was at a lower scale" (p. 7). He suggested that universities should adopt a more comprehensive approach in dealing with the problem of plagiarism and should address the fundamental reasons why students plagiarize.

Rama (2019) explored the scholarly views on different aspects of plagiarism and its prevention and found positive impact of use of detection softwares on controlling plagiarism despite of some major limitations and misconceptions about the technology. Similarly, Meo and Talhal (2019) concluded that plagiarism detection software are basically text matching softwares and cannot be solely relied for plagiarism prevention. Rama (2019) recommended introduction of anti-plagiarism courses in the curriculum and believed that librarians along with faculty can play an important role in creating awareness and developing skills against plagiarism. Perkins *et al.* (2020) believed that the lack of writing skills is an important cause of plagiarism incidents in Vietnam universities and explored the use of detection software as a tool to assist in detecting contract cheating by matching earlier writing samples of the students as "fingerprints". They discussed an intervention design, Academic English Masterclass (AEM), to improve the English writing ability of the student through provision of additional learning courses and submitted work under this program will be treated as fingerprints. They found that the level of contract cheating has decreased in the cases where intervention has been provided; however, they also accept that design has its own limitations. Therefore, they stressed that a holistic approach of intervention is required to prevent plagiarism.

Mostofa *et al.* (2021) explored statistically the impact of plagiarism detection tools on the level of plagiarism awareness among researchers of University of Dhaka, Bangladesh and found a high level of awareness among researchers on plagiarism issues. They stressed to promote the use of detection tools along with special trainings on plagiarism avoidance.

4. Research objectives

The study aims to explore the perceptions of university librarians on the following:

- the reasons of unethical use of academic literature by university scholars in their academic writings; and
- the effectiveness of Turnitin software in deterring plagiarism and other strategies to prevent plagiarism.

5. Methodology

This paper is based on partial data obtained through semi-structured interviews conducted for the doctoral dissertation of the first author. The design of the study was qualitative and survey research method was used to achieve objectives of the study. Interviews were conducted to explore the causes of unethical practices among Pakistani graduate scholars. Previous studies presented the perceptions of students and faculty on the causes of plagiarism. The views of other stakeholders like research facilitators (librarians) involved in plagiarism prevention practices were not explored in South Asian context. Therefore, nine library heads of well-established central university libraries in Pakistan were interviewed to get their views on the causes of unethical practices among Pakistani graduate scholars, with specific reference to South Asian academic culture and the ways plagiarism can be controlled. The selection of this interviewees' group was purposive and purely based on the findings of a prior nation-wide survey conducted for the PhD study to explore the prevailing anti-plagiarism practices in Pakistani universities and role of university libraries in its prevention.

The group of interviewees was highlighted in that survey as the fourth major group administrating similarity index check using "Turnitin" software service provided by Higher Education Commission, Pakistan to check the PhD dissertations for plagiarism before final submission. Due to this work assignment, those heads had a close interaction with the university graduate scholars along with their supervisors and had views on the issues of plagiarism and research integrity based on their direct observation. Moreover, these heads were also familiar with the HEC's plagiarism policies and have practical understanding of plagiarism due to their daily interaction with the matters and comprehension of HEC's anti-plagiarism booklets. An interview guide was used to ask questions. Five out of the nine library heads were (stationed at Lahore city, the city of the interviewer) interviewed in their offices, while telephonic interviews were conducted situated in other cities (two from Islamabad and one each from the cities of Karachi and Topi).

A mobile voice and call recording system were used with the permission of the interviewees to record the interviews. During conduct of interview to avoid misunderstandings and communication gap both languages, English and Urdu were used. The duration of the interviews varied from 15 to 45 min. The text was transcribed and thematically analysed manually to extract and organize the findings related to the causes and deterrents.

However, to maintain the anonymity of the nine interviewees, a reference code was used, i.e. P1 (Participant 1), P2 (Participant 2), etc.

6. Findings of data analysis

6.1 Perceived causes of plagiarism

The interviewees were asked to share their views on the hindrances in the effective plagiarism prevention among scholars with special reference to Pakistan. Their views indicate four primary areas of concerns (Figure 1).

6.1.1 Lack of skills. All interviewees indicated a lack of various academic skills among researchers caused stress in the researchers and that further caused incidents of both

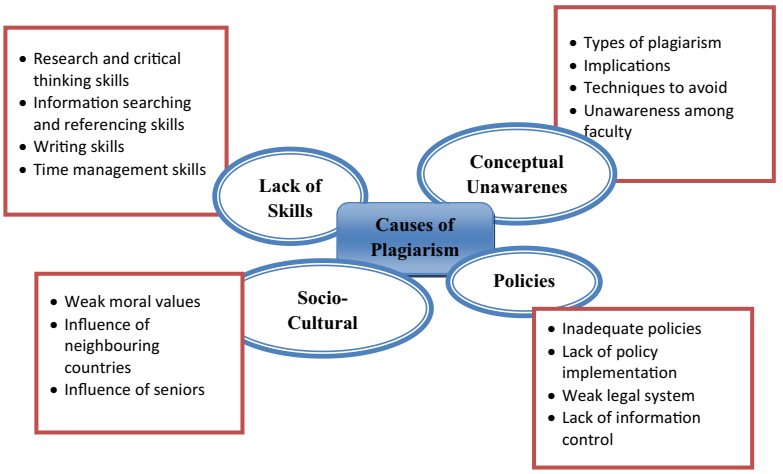


Figure 1.
Perceived causes of plagiarism

intentional and unintentional plagiarism. The deficiency of the following skills among scholars was mentioned by the interviewees.

Five out of nine interviewees believed that lack of research training increases academic integrity violation incidents. They mentioned that despite mandatory teaching of research courses in MPhil and PhD programs, the researchers were not trained to practically use this knowledge in actual research projects. As indicated by one interviewee (P1):

> *Lack of training to conduct a research project according to required standards is a big problem in plagiarism avoidance. There is no concept [teaching] of research at the undergraduate level in our education system. Therefore, we start with teaching theories of research at the post-graduate level, rather than actually practicing these theories at that level.*

Another interviewee (P7) claimed that real efforts to instruct research methodology in the academic institutions starts from post-graduate level; therefore, the plagiarism awareness level among the undergraduates is very low. Provision of continued research training programs from the universities for enhancing students' research skills was demanded by many interviewees. The interviewee (P8) believed that students do not learn during workshops and lectures but they learn when they practice it afterward. Hence, there should be some continued training arrangements.

Lack of uniformity prevailing in education systems in Pakistan was criticized by many interviewees as they believed that this causes variation in policy implementation. The library head of a private sector university (P5) believed that traditional matriculation system is the main reason for plagiarism in Pakistan as this system lacks creative skills development among students and support rote learning. Maxwell *et al.* (2008) explored the cultural influences on the general understanding and perceived seriousness of plagiarism on local and Asian students studying in Australian universities. They reported that learning through memorization is still a major technique used in modern Asian classrooms. In their view, this method of learning might have resulted in learned plagiarism and lack of understanding of the Western ideas of textual ownership.

Inability to handle information sources or lack of information skills, i.e. searching, organizing and reusing of information was perceived as a major source to cause plagiarism. The interviewees indicated that a scholar's inability to find the required piece of information

increases his/her over-reliance on unauthentic, controlled information resources such as Google. Moreover, the disappointment of a researcher in finding information causes stress and increases temptation for plagiarism. An interviewee (P3) mentioned that lack of scholars' acquaintance with reference manuals boosted the incidents of unintentional plagiarism. The students collect incomplete bibliographic information of a source due to lack of guidance and practice. Carnero (2017) expressed similar views and stated that for ESL students learning citation skills is a big challenge and minimizes the student ability to avoid plagiarism.

Lack of English language skills believed as a fundamental element in plagiarism prevalence among Pakistani scholars. Six out of nine interviewees showed their belief that despite having sufficient knowledge in their discipline the Asian scholars lack language proficiency. These scholars mostly fail to script down their knowledge in a second language with the required academic style in a limited time frame. Therefore, they prefer to mix up the writing chunks of other scholars with their original research data to meet the academic requirements.

The interviewee (P7) mentioned that lack of paraphrasing skills prevails not only among students but also even some teachers are not familiar with paraphrasing skills. Hence, most of the scholars cannot use the information sources that support their arguments in an appropriate way. Studies conducted on ESL students (Hayes and Introna, 2005; McGowan, 2008; Stapleton, 2012; Sutherland-Smith, 2005) also highlighted the lack of English language skills as a critical reason of plagiarism in Asian students.

Two interviewees (P1, P7) mentioned the importance of time management skills in controlling plagiarism among scholars. The interviewees stated that a majority of students were not trained to work hard and to complete the tasks by deadlines. However, these interviewees perceived that the skill of time management was associated more with social lifestyles than academic training. Various studies (Nushi and Firoozkohi, 2017; Ramzan *et al.*, 2012; Shirazi *et al.*, 2010) indicated that lack of time management skill was an important element in prevalence of unethical writing practices in Asian scholars.

6.1.2 Conceptual unawareness about plagiarism. Conceptual unawareness about plagiarism was highlighted by seven out of nine interviewees as a prime concern to plagiarism control. The interviewees described that unawareness was based on ignorance of what plagiarism is; its implications; and its avoidance. One interviewee (P1) shared the findings of his own study on plagiarism awareness among university students in a Pakistani university and found that 23% students were completely unaware with conceptual understanding of plagiarism. He emphasized that awareness with the consequences of plagiarism is necessary for the researcher to avoid plagiarism as unawareness with plagiarism implications creates non-serious attitude towards the issue. He further disclosed that during a lecture series, he delivered all over the country on awareness about plagiarism, he observed that even some senior academic individuals like Deans of certain departments, had no conceptual understanding of plagiarism and were not familiar with the international research ethics.

His stance of conceptual unawareness of plagiarism was further supported by another interviewee (P8). She (P8) lamented that in some cases research supervisors do not have awareness with report writing standards and in case if they have awareness, they simply do not instruct their students properly. Interviewee (P4) stated that small scale private universities are hiring young faculty members, usually graduated from their own institution and do not have sufficient experience of research and its ethics. Moreover, the senior faculty is burdened with the research related work and they find it hard to give needed time to each student.

Lack of awareness and training on research ethics, faculty unawareness regarding conceptual understanding of plagiarism has been highlighted by some international and Asian scholars as well. Erkaya (2009) while exploring the causes of plagiarism in college students found that the students believed that their teachers were not properly trained to teach them how to write. Starovoytova and Namango (2016) in an empirical study found lack of understanding on the basic elements of scientific writing and plagiarism among engineering faculty in a university of Kenya.

Shirazi *et al.* (2010) identified the lack of conceptual awareness of plagiarism among Pakistani medical faculty members as well. Punyani and Deshpande (2018) found a lacuna regarding the authors' understanding of important basic concepts in scientific writing that includes the concept of plagiarism. Nushi and Firoozkohi (2017) found while assessing the role of an Iranian university TEFL (Teaching English as a Foreign Language) program teachers on imparting plagiarism understanding among their students, that only 16.4% syllabuses which they examined, included a clear plagiarism policy. They concluded that "students receive little instruction or information on plagiarism and plagiaristic practices from their teachers and universities". Faculty has two important roles in the campaign of academic integrity promotion as prime anti-plagiarism instructor and motivational role models. Therefore, Auer and Krupar (2001) believed that unawareness among faculty regarding policies and implications of plagiarism cause more harm to academic integrity than unawareness among students.

Mansoor and Ameen (2016) mentioned that in most of the universities "advice in class" was the prime method to create awareness against plagiarism among researchers. This advice is usually based on the knowledge of the teacher. However, as a couple of interviewees indicated that the faculty's own knowledge of ethical and unethical practices was not sufficient, hence, the reliance on this method becomes questionable. It appears that the ethical academic practices awareness and training campaign in Asian universities must be initiated and arranged for all academic stakeholders including faculty, as suggested by other studies conducted in Pakistan (Shirazi *et al.*, 2010; Ramzan *et al.*, 2012), India (Jayaraman, 2012; Kumari *et al.*, 2018) and Iran (Nushi and Firoozkohi, 2017).

6.1.3 Socio-cultural factors. The interviewees believed that socio-cultural factors influence on researchers play an important role in crafting a non-serious attitude towards plagiarism. Three interviewees stated that overall moral values of the societies are deteriorating. They believed that the students who observe and experience moral poverty at an early stage of their life generally practice that at the later stages of their lives. They stated that in the subcontinent culture, the owning of others' work is not generally considered as unethical.

The interviewee (P3) believed that sometimes students learn to behave unethically due to some cultural influences. He believed that when the social values of integrity, hardwork and fairness lack in a society people find it easy to indulge in unethical practices. Hence, to gain financial benefits, quality of work is compromised. Two interviewees considered that the unethical practices prevailing in the neighbouring countries provided bad examples and negative motivation for the students. They mentioned that in this region some countries were openly making copies of the industrial designs of other countries and under this scenario it became difficult to ethically motivate the students. The interviewee (P7) argued that Pakistani students take plagiarism as culture of Asian countries, when they see that the publishing industry of country in their immediate neighbour is printing foreign authors' books with the names of fictitious local authors.

Batane (2010) pointed out the negative impact of surrounding environment on a research plagiarism perception. He believed that the act of plagiarism is a behaviour sometimes

acquired through observing and following peers and senior colleagues who successfully advanced in their academic endeavours without suffering any negative consequences from plagiarism. Harris (2001) stated that sometimes students give excuses that people at role models positions do all these unethical deeds before becoming experts (p. 6).To correct these doctrines based on assumptions and false interpretations, it is necessary to develop an academic research culture where researchers are well aware of the negative implications of plagiarism on society and individuals and take promotion of academic integrity as their personal moral obligation.

6.1.4 Administrative problems. The interviewees indicated that administrative practices and policies of higher education institutions negatively influence academic integrity fabrication. Seven interviewees believed that HEC's policies regarding making faculty promotions negatively resulted for research quality and integrity. Three interviewees (P1, P3, P7) pointed out that HEC's policy of making faculty promotions dependent on publication of a certain number of research papers in internationally recognized journals, placed a negative impact on the researchers. The policy increases the stress due to time pressures. The "publish or perish" policy has increased peer pressure among faculty members. The interviewee (P1) pointed out that without providing proper training and mechanism to faculty the implementation of foreign standards for academic promotion has not promoted quality research culture rather creates a stressful race to save the job or to earn more. Therefore, the benefits of research for the relevant industry are very limited due to the low quality. Elliott *et al.* (2013) believed that:

> If time demands can cause students to cheat, time demands on faculty could be one of the factors leading to faculty plagiarism and fraud [. . .] they feel pressured by the tenure clock and/or negative performance evaluations (p. 95).

Lack of effective mechanisms for implementation of plagiarism policies in institutions was another element mentioned by the interviewees (P1, P7). They believed that the problem increases when the implementation level of policies and conceptual understanding about plagiarism varies from university to university and from teacher to teacher. Lack of policy consistency confuses the researchers and provides outflows for the academic integrity offenders. Interviewee (P1) stated that common student perceptions that "nobody catches me" and "nobody cares" emerge from lack coordination, implementation and consistency of policies. Ramzan *et al.* (2012) also found in their study that graduate and postgraduate students "do not believe that existing plagiarism policies are effective in detecting the plagiarism and consequently awarding punishments to those who violate the plagiarism policies".

The ambiguities in defining of plagiarism and fair use coupled with the legal system are also of concern. Interviewee (P2) shared an incident of a student at his institution. The student filed a case against university administration when he was given a harsher penalty on the second attempt of plagiarism. The student got relief by the court under a Pakistani law that if a person is previously awarded a lesser penalty for a misconduct cannot be punished with a harsher penalty in future for the same crime. The interviewee commented that the policymakers at the national level should address legal loopholes in this regard.

The views of the interviewees indicate the need for revision of current policies, and inclusion of such measures that would help to create an environment where plagiarism could be stopped before it happens. Furthermore, simply adopting policies and applying rules without providing the prerequisites to avoid plagiarism would only produce superficial cosmetic results.

The above aspects mentioned by the interviewees indicate the major causes of plagiarism in Asian cultures. Their views endorse more or less the findings of the previous studies while adding to some indigenous issues.

6.2 Deterrents to plagiarism

The interviewees were also asked to share their experiences and perspectives on effective plagiarism deterrent strategies. Their views indicated positive and negative impacts of the prevailing plagiarism detection strategy in Pakistani universities along with the prospects of a much-desired guidance strategy to meet the skill deficiencies among the researchers (Figure 2).

6.2.1 Plagiarism detection software. The interviewees believed that the use of Turnitin software had not only positively reduced the ratio of plagiarism incidents but also helped in highlighting the act of plagiarism as an academic crime in Pakistan. They perceived that the use of software created more awareness among postgraduate students as compared to graduate and undergraduate students. Interviewee (P2) commented:

> *Plagiarism was always in practice but its detection with proof was not easy. Turnitin has played a vital role not only in the control of plagiarism but it also highlighted plagiarism as a major issue for academic integrity. Although the software has limitations, overall, it proves a good source to control plagiarism.*

The experiences shared by the interviewees, however, indicated that due to certain limitations of the software it cannot be solely relied to prevent plagiarism. The interviewees mentioned that although the software was good to detect plagiarism among English language research, it does not deal with the research published in regional languages such as Urdu, Punjabi, Sindhi and oriental languages, i.e. Arabic and Persian. Similarly, the software is unable to check information resources that remained unpublished, published but not digitalized, and digitalized but not included in any database recognizable to Turnitin. They found that students were using techniques to cheat the software and due to this the deterrent effect o1f the software is reduced. Interviewee (P6) stated that use of Turnitin

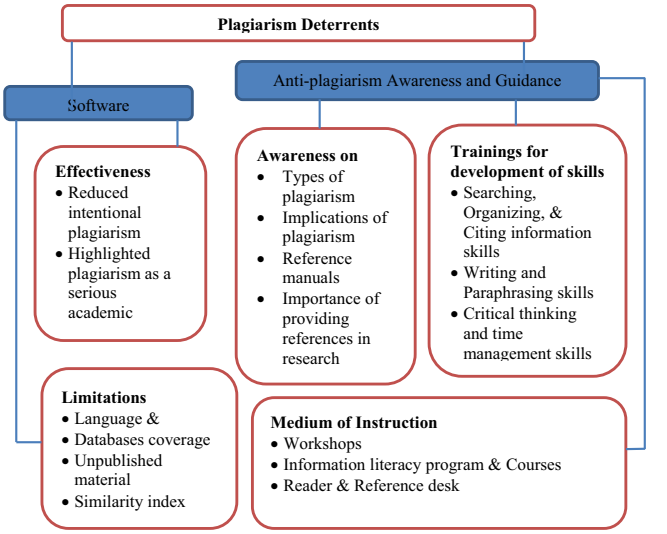


Figure 2.
Perceptions on plagiarism deterrents

software initially created a deterrent effect, but with the passage of time, people find out techniques to cheat it, and now, its effectiveness has reduced.

It is difficult to detect intentional plagiarism with the Turnitin as the software can only detect exact similarity between two documents. Furthermore, in certain situations the highlighted paragraphs in a similarity index report does not mean plagiarism. Interviewee (P3) shared that:

> A researcher of our university submitted an article work for Turnitin check. To my surprise, Turnitin showed no plagiarism in the article while the text of that (article) was plagiarized from a thesis available on an internet database. Actually, I was aware with the ability of that researcher and his work was not matching with his calibre. I put some phrases from his article simply in Google search and as I assumed the material was plagiarized from a thesis of neighbouring country's university.

Similar views regarding Turnitin limitations were also mentioned (Hayes and Introna, 2005; Stapleton, 2012). Interviewee (P7) believed that the role of the supervisor is indispensable to evaluate the research work of a student, and the similarity index report can helpful for this purpose. Hence, mere results of a similarity report should not be taken equal to a final proof of an instance of plagiarism.

6.2.2 Anti-plagiarism awareness and guidance. The majority of the interviewees believed that conceptual unawareness and lack of various academic skills regarding plagiarism were major causes of plagiarism in Pakistan. They suggested that the provision of anti-plagiarism awareness and guidance among university academia is an effective strategy to counter plagiarism. They further stressed that university libraries being the research centre could play an important role in this anti-plagiarism awareness and guidance of various stake holders. They believed that anti plagiarism awareness would urge the researcher to make personal efforts to learn anti-plagiarism skills. Interviewee (P9) claimed that researchers mostly do not realize the importance to give references. He suggested that during plagiarism awareness programs the students should be informed about right way of citing authors is necessary to validate their argument in literature review section.

Three interviewees believed that efforts of a library with short term workshops and training programs would not bring the desired results. They recommended that a compulsory course on information ethics should be taught at undergraduate and postgraduate levels with hands-on practice for information handling tools, i.e. citation software. They also added that information professionals can become instructor of these courses. The interviewees supported the views of the scholars (Auer and Krupar, 2001; Gibson and Chester-Fangman, 2011; Hornreich, 2011; Kloda and Nicholson, 2007; Mansoor and Ameen, 2018) and agreed that university libraries can play a vital role in preventing plagiarism and promoting academic ethics by providing such workshops. It will help to develop research-writing and referencing skills of the researchers. They insisted that plagiarism detection software should be used as a follow up phase of an extensive anti-plagiarism awareness and guidance strategy of the universities and not the only way of deterrent (Batane, 2010; Stapleton, 2012).

7. Conclusions and recommendations

The interviewees believed that lack of conceptual understanding and information skills along with the implications of socio-cultural environment and irregular policies are the prime reasons of Pakistani scholars' attempts of unethical scholarly behaviours and plagiarism. The views of the administrators of a plagiarism detection software service in

Pakistan corroborate that we need to do more than just relying on digital technology for determining plagiarism. The study findings indicate the need for revision of current policies and inclusion of measures that would help to create an environment, where scholars are sensitive about information and research ethics to avoid plagiarism. It is evident that merely adopting policies and applying rules without providing the prerequisites to avoid plagiarism would only produce superficial and cosmetic results. On the basis of the experiences and suggestions shared by the interviewees, the following recommendations are made to deter the unethical use of information among researchers:

- Universities should consider further strengthening of their prevailing strategy to control plagiarism and take steps to ensure the provision of anti-plagiarism guidance, training for researchers and the service of plagiarism detection software.
- The revised policies should focus on inducing strong ethical values for information usage among students at an early academic age along with other measures. Anti-plagiarism awareness should be a part of the secondary and undergraduate syllabuses. At higher level, continuous guidance and training on research ethics should be provided through seminars, workshops and credit courses at the graduate and postgraduate levels in universities.
- Uniformity and consistency in the implementation of anti-plagiarism policies and practices are indispensable to prevent plagiarism.
- The libraries can play an active role in facilitating anti-plagiarism guidance to researchers. They must be included in the universities' anti-plagiarism campaigns and participate in the activities against plagiarism by their respective administrations.
- Special attention should be given to create awareness among researchers on the institutional plagiarism policies through websites and handbooks.
- The use of software as a formative feedback and writing tool should be encouraged at the early learning stage of the research programs. The login accounts with instructor's rights in Turnitin's software should not be assigned or given out to students.
- Last but not least, imparting the academic and scholarly writing skills is utmost important to make them confident writers.

8. Future research

The findings of the study corroborate the results of the many studies conducted to explore causes of plagiarism with Pakistani scholars' perspective along with indicating some new areas of concern. The study highlights the need to further explore the subject with associated dimensions, i.e. studies to assess the:

- Level of conceptual awareness of Asian faculty about plagiarism and their instruction methods to deliver advice against plagiarism in the classroom.
- Vulnerabilities of Asian pedagogical and socio-cultural practices in relation to research misconduct.
- Prospects of Turnitin Software as writing and learning tool in the Asian academic culture.
- Level of conceptual awareness of university librarians about plagiarism and their actual skills' capacity to play the role of an anti-plagiarism instructor.

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